

What is Seal actually for?

Prompted by a fair question: why build this when iMessage and Signal already exist? Below is the honest comparison, including where Seal loses.

1 The comparison

The row that matters is the first one. Everything else follows from it.

	SEAL	IMESSAGE	SIGNAL	WHATSAPP	WORLD ID
What it proves	You two physically met Signed, timestamped record of a real encounter	You control a device Apple ID owns the binding	You control a key Safety numbers verify the key, not the person	You control a phone number	You are a human Not which human, and not that you know anyone
End-to-end encrypted	Yes AES-GCM, ratcheted sender keys	Yes PQ3 post-quantum since 2024	Yes The protocol others license	Yes Signal protocol, under Meta	N/A — not a messenger
Identity root	FIDO2 key or passkey Yours. No account to seize	Apple ID	Phone number + key	Phone number	Iris scan at a physical Orb
Phone number required	No	Apple ID (email ok)	Yes, to register Username hides it; doesn't remove it	Yes	No
Can a stranger reach you	Structurally impossible No graph path = no channel. The real feature	Yes Anyone with your number or email	Yes Anyone with your number or username	Yes	N/A
Cost to fake one account	An in-person meeting Doesn't scale for an attacker — the whole point	A device	A burner SIM	A burner SIM	A trip to an Orb
Central database of you	None Signatures held by users; display names public	Apple holds the graph	Minimal Sealed sender; nonprofit	Meta holds the graph	Biometric template held centrally
Deniability	None — signed by design Liability for a source; asset for compliance	Partial	Deliberate You can deny you sent it	Partial	N/A
Lose your credential	Identity is gone Backup keys still unbuilt (FR-3)	Apple recovery	PIN recovery	SMS recovery	Key rotation + recovery, 2026
Platforms	iOS 26+ only Needs iCloud. No Android, web, or desktop	Apple only	Everywhere	Everywhere	SDK, cross-platform
People on it	You and your mom Architecture ceiling ~1k identities today	~1 billion+	~100 million+	~2 billion+	450M+ verifications claimed
Server cost to run	≈ \$0 CloudKit + one static file	Apple's problem	Donations	Meta's problem	VC-funded

2 The honest verdict

For your family group chat, Seal is worse than iMessage — and it always will be.

Your mom is already reachable by everyone she loves on an app that works on every device, backs up automatically, never loses her identity, and requires no ceremony. Seal asks her to do more work for a benefit she doesn't feel, because she was never going to get scammed by someone in her family group chat.

That isn't a flaw in Seal. It means **the family group chat is the wrong yardstick**. Every feature you'd add to win that comparison — SMS fallback, easy recovery, cross-platform, frictionless adding — destroys the one property that makes Seal different.

3 What Seal has that nobody else does

Read down the first row of the table again. The three products split cleanly:

- **Signal and iMessage prove a key.** They secure the *pipe*. Neither has any opinion about whether a human is on the other end — exactly the problem that got worse every year since 2023.
- **World ID proves a human.** But an anonymous one. It can tell Tinder "not a bot." It cannot tell you "this is the person you met at Sarah's wedding," and it requires handing your iris to a company.
- **Seal proves a relationship.** Not that you exist — that *you two met*. Nobody else is building this, because it can only be built as a social graph, and a social graph is the hardest thing to bootstrap.

The strongest asset

The forge log. A signed, years-long record of every human you've actually met. It compounds, can't be transferred, and can't be reconstructed if you leave. A real moat — and not a messaging feature at all.

The strongest feature

Strangers have no channel to you. Not filtered, not spam-scored — structurally absent. As AI-driven scams get cheaper, "unreachable by default" gets more valuable every year.

The hardest problem

Cold-start. Value is zero until your friends are on it, and the ceremony makes each addition cost a real meeting. Every network faces this; yours has the steepest version by design.

The quiet risk

"Zero servers" is really "Apple's servers." CloudKit means iOS-only forever, and Apple sets the ceiling on who can ever join.

4 The question worth answering next

Not "why is this better than Signal" — it isn't, and shouldn't try to be. The question is:

Who has a problem that only "prove we met" solves — and feels it this month?

Candidates, roughly in order of pain: dating safety before a first meet · marketplace and rental handoffs between strangers · conference and meetup attendee graphs · elderly relatives targeted by impersonation scams · high-value transaction confirmation, where "is this really my client" now costs companies real money.

Your vision doc already names the event loop and the Square. Those express the wedge. A family group chat does not — it's a fine testbed for the crypto, but it will never make the case for the product, and it will always lose the comparison your mom just made.

Sources: Apple Security Research — iMessage PQ3 & Contact Key Verification · Signal Support — Phone Number Privacy & Usernames · World ID expansion announcement, 17 April 2026 · Seal figures from docs/VISION.md, HANDOFF.md, and the codebase. User counts for iMessage, Signal, and WhatsApp are widely-reported approximations, not audited figures.